

Cade Tanaka

Seattle, WA | +1-206-683-7607 | cadetanaka@gmail.com | [LinkedIn](#) | [GitHub](#)

Education

University of Washington, Seattle, WA

Aug 2025

BS, Computer Science and Mathematics

- **GPA:** 3.61
- **Achievements:** Deans list, 2021-2025
- **Coursework:** Software Design and Implementation, Data Structures and Parallelism, AI, Hardware/Software Interface, Foundations of Computing and Graph Theory, Database Systems, Statistics/Probability, Linear Algebra and Differential Equations

Work Experience

Tanaka Lawn Care, LLC

Jul 2020 - Aug 2025

Software Developer and Co-Founder

- Established and grew a lawn care business, combining technology and efficient processes to deliver affordable services to clients in need.
- Built a user-friendly website that drove a 60% increase in client engagement and supported \$200,000 in total revenue.
- Implemented Dijkstra's and Bellman-Ford algorithms to optimize travel routes, reducing travel times by 14% and enabling service to 12 yards daily.
- Oversaw team operations, aligning project goals with deadlines and budget constraints to deliver consistent results.

Projects

AI Board Game Bot (Python) 

Feb 2025 - Jun 2025

- Designed and implemented a Connect Three bot using minimax algorithm with Alpha-Beta pruning, achieving move calculations in under 0.1 seconds across three game variations.
- Ranked top 5 out of 200+ students in class competition by optimizing bot performance for high-depth decision-making.

Cargo Packing for SDS Company (Python) 

Sep 2024 - Jan 2025

- Collaborated with SDS, a Chinese shipping company, to optimize container packing processes, reducing wasted space to less than 10%.
- Designed detailed packing diagrams to guide workers while ensuring adherence to all packing requirements.
- Created visual models to validate and showcase the solution's efficiency and practicality.

Chess Engine (Java + React) 

Mar 2024 - Feb 2025

- Engineered a fully functional chess AI using the minimax algorithm, optimizing decision-making with depth-based search techniques.
- Utilized advanced tree pruning techniques to improve chess AI efficiency, maintaining a 0.4-second decision time at a depth of 5.
- Designed and implemented an interactive UI in React, allowing users to visualize possible moves directly on the chessboard.

Hospital schedule interface (SQL) 

Apr 2023 - Jul 2023

- Developed a hospital scheduling interface that streamlined patient account creation and appointment booking.
- Optimized patient record storage by integrating an SQL database, reducing access times by 12% while ensuring data security.
- Implemented a secure username and password system, safeguarding confidential data for over 4,000 users.

Campus guidance map (Java + React) 

Sep 2023 - Nov 2023

- Developed an interactive campus navigation tool using Dijkstra's algorithm to calculate and display the shortest path between buildings.
- Implemented an intuitive React UI for campus navigation, allowing users to input building locations and visualize results on a map.
- Designed a map visualization feature to showcase the shortest path between buildings, improving navigation accuracy.

Technical Skills

- **Programming Languages:** Java, Python, C, SQL, R, Scheme, OCaml, Ruby, C++
- **Web Technologies:** React, HTML
- **Frameworks and Tools:** Git, Azure, IntelliJ, React, Angular